Remote Sensing Change Detection On Land Using HRNet

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***Abstract*—Timely detection of changes such as deforestation, urban sprawl, or shifts in land use is crucial in formed decision making and policy formulation. Additionally, in disaster scenarios like floods or wildfires, rapid change detection can guide emergency response and resource allocation. Change detection is used for identifying the changes in an object or phenomena over time.Traditional pixel-based change detection often struggles with complex, non-linear patterns in large-scale remote sensing data.In this work, we propose a modified High-Resolution Network (HRNet) framework for bi-temporal remote sensing change detection. Unlike prior Siamese HRNet , our model emphasizes efficient preprocessing, visualization through a custom GUI, and benchmarking against both classical CNNs and modern architectures. HR networks, which are capable of learning hierarchical representations of spatial and temporal data. This introduces a method using HR networks for change detection in remote sensing on bi temporal data, and it is highlighted through their potential to analyze bi-temporal aerial or satellite images to detect fine and large changes using SEmantic Change detectiON Dataset(SECOND dataset) in land use, infrastructure, and other environmental factors. Our proposed methodology uses HR networks to give better performance. We trained and evaluated our model on the publicly available SECOND dataset that consists of 4,662 bi-temporal image pairs from urban regions.**

Keywords—HR Networks, Remote sensing, Spatial and temporal data, Change detection, satellite images ,aerial images, non linear patterns .

I. Introduction

Remote sensing change detection plays a vital role in monitoring land-use and land-cover dynamics, including urban expansion, vegetation growth or loss, and changes in water bodies. Accurate detection of such changes supports sustainable urban planning, environmental protection, and disaster response . Traditional pixel-based approaches often fail to capture complex non-linear patterns in high-resolution imagery. Although deep learning models such as U-Net, SegNet, and DeepLabV3+ have improved performance, they suffer from loss of spatial details due to repeated downsampling. Moreover, existing Siamese HRNet and HRNet-Transformer models achieve high accuracy but at the cost of increased complexity and computational demand, limiting their practical deployment.We apply a deep learning model called HRNet (High-Resolution Network) to be able to identify the changes reliably and accurately. The goal of this project is to apply HRNet to design more intelligent, rapid, and effective land monitoring, thus making it easier for decision-making.

Change detection in remote sensing is defined as the identification of changes in the land cover over time in bi-temporal satellite imagery. By comparing two different time periods of imagery (2006 and 2012), remote sensing change detection enables us to identify what areas have changed due to human or natural activity.

High-resolution networks or HRNets are a type of deep-learning architecture that aim to enable high-resolution feature maps, which are useful in image segmentation, pose estimation, and other computer vision tasks. HRNet maintains high-resolution representations through the whole network, which means unnecessary downsampling is avoided.

Remote sensing is the ability to evaluate an object or area from a distance, utilizing remote technology to take the measurement. Remote sensing technology uses the satellites, aircraft, drones, and ground-based sensors or platforms.

The key contributions of this paper are as follows:

1. We propose a streamlined HRNet framework for bi-temporal remote sensing change detection, avoiding the complexity of Siamese or hybrid transformer architectures.
2. We present an efficient preprocessing pipeline, including RGB-to-class mapping and ignore-index handling for noisy pixels.
3. We design a visualization module and GUI for pixel-level analysis, making change maps interpretable for non-technical users.
4. We conduct extensive experiments on the SECOND dataset, reporting both overall metrics and per-class performance, and compare against CNN-based and modern transformer-based baselines.

This document presents the introduction in Section-I of the project, the Literature Surveys in Section-II,and it is followed by Section-III which describes the methodology which has been done in our research study, Section-IV says about the results obtained, Section-V has conclusion and the future work of the research work and the final section has references which are used in this document.

II. Related work

L. Fazry et al [1] propose Change Detection of High-Resolution Remote Sensing Images using Adaptive Focal Modulation on Hierarchical Feature Maps. They introduce FocalCD, a change detection method that learns both long and short dependencies. FocalCD also has efficient multi-scale feature fusion, as well as a pyramidal

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decoder. FocalCD achieves F1 scores of 0.9851, 0.952, and 0.9616 on the CDD,LEVIR-CD,WHU-CD datasets, respectively change detection models while having comparable or lower computation complexity. The main disadvantages are higher training complexity and it has limited generalization to other change detection tasks.

Zihao Liu et al. [2] presented a change detection method for high-resolution remote sensing images using a UNet-based architecture, where ResNet50 is used in the encoder to extract deeper features and reduce training time, and the scSE attention module is added in skip connections. The method showed improved precision, recall, and F1 scores on LEVIR-CD and CDD datasets, with a model size of 5.37M parameters, performing better than many existing CD networks. However, it has a few drawbacks, the model tends to struggle when trained on data with noisy labels, and it also demands more computational power.

Huayu Zhang et al. [3] presented a change detection method for high-resolution remote sensing images using a Siamese Holistically-guided FCN (SHG-FCN) model, designed to extract both low and high level features for accurate change detection. The model includes a Siamese encoder and a holistically-guided decoder, and it performed effectively on two public HRRS datasets. While the results are promising, the method has a few drawbacks as the Siamese structure leads to higher computational cost, making the model heavy to run.

Peng Tao et al. [4] introduced a HRNet-Transformer Network Integrating Recurrent-Tokens for Remote Sensing Image Change Detection. They used HRNet to preserve the spatial and channel information in image features and to avoid the loss of spatial information caused by convolution-based methods. Their model outputs a binary mask map with consideration to both pixel classification and location. The disadvantages are that the architecture integration of HRNet and Transformer components into a single model architecture makes it more challenging to design and train a single model architecture.

Yu Wang et al. [5] presented a Siamese High Resolution Network for change detection, using HRNet to generate accurate and detailed results. The model achieved precision, recall, and F1 scores of 0.942, 0.922, and 0.932 on the CDD dataset. However, it has some disadvantages as the model is large and complex, demanding more computational resources, and its performance may drop when applied to datasets with high variability.

Zhi Cheng et al. [6] presented a Siamese remote sensing image segmentation method based on HRNet, which maintains high resolution feature maps and outperforms traditional methods. The model combines DICE loss and BCE loss to address sample imbalance, showing better performance than U-Net and DeepLabv3+ with mPA of 0.9545 and mIOU of 0.8141. However, it has some drawbacks as the architecture is complex and harder to implement, and the training time is longer due to high resolution processing and model structure.

T Chen et al. [7] presented a Siamese remote sensing image segmentation method based on HRNet, which integrates multi-dimensional features and uses a tensely upsampling method to enhance feature extraction while reducing information loss. The model achieved strong results on the Change Detection Dataset, with a precision, recall, and F1 score of 0.953. Despite its effectiveness, the method has some drawbacks such as a complex and resource-intensive structure, and its performance may degrade when the input data is noisy or of low quality.

Fei Pan et.al. [8] developed a Temporal-Reliable Method for Change Detection in High-Resolution Bi-Temporal Remote Sensing Images. The paper proposes a Temporal-Reliable Change Detection (TRCD) framework, which uses the improved HRNet architecture for high-resolution remote sensing image change detection. They achieved a precision,recall,F1score of 94.60,88.13, 91.25 with an overall accuracy of 97.91 when using 6000 pairs of images. However, when using the LEVIR-CD dataset consisting of 4096 pairs of images, the respective precision, recall, and F1 scores were 66.74, 45.22, and 53.91. The disadvantages are the it increases model complexity, Model performance can differ drastically across datasets.

R Caye Daudt et al.[9] (2018) proposed Fully Convolutional Siamese Networks (FC-Siamese) for pixel-wise change detection in bi-temporal images. The method performed well on benchmarks but is limited by dependence on labeled data and generalization challenges.

X. Zhang et al.[10] (2019) introduced a framework combining Stacked Denoising Autoencoders (SDAE) with a semi-supervised Chan–Vese (SCV) model for land cover change detection. While effective, it faces generalization issues and high computational cost.

L. Ding et al. [11] (2022) proposed a bi-temporal semantic reasoning approach for change detection in high-resolution remote sensing images, achieving improved accuracy but requiring substantial labeled data and higher computational resources.

P. Zhang et al.[12] (2018) developed ASPP-Unet and ResASPP-Unet for urban land cover classification from WorldView imagery, achieving strong accuracy. However, the models are less flexible across regions and computationally demanding.

Bai et al. [13] (2022) developed a comprehensive review of deep learning approaches for change detection, covering architectures, datasets, challenges, and future directions in remote sensing.

Wang et al. [14] (2023) proposed Double U-Net (W-Net), a dual-head architecture for remote sensing change detection, enhancing feature representation and boundary precision.

Jiang et al. [15] (2022) surveyed deep learning-based change detection methods for high-resolution images, highlighting strengths, limitations, and research trends.

III. METHODOLOGY

The proposed system makes use of a modified High-Resolution Network (HRNet) architecture and a deep learning technique for remote sensing change detection. Data preparation, model architecture design, training, evaluation, and visualization of results are the steps in the methodology. Every step is carefully designed for the precise identification and categorization of changes in land cover from bi-temporal satellite images.

A. Data Preparation

The SECOND (Semantic ChangE detectiON Dataset) focuses on remote sensing and is meant for pixel-wise precision in change detection. It has 4,662 pairs of high-resolution aerial images from diverse urban regions of Hangzhou, Chengdu, and Shanghai.The dataset works with six land-cover classes: non-vegetated ground surface, tree, low vegetation, water, building, and playground.To improve generalization, image augmentations such as random horizontal and vertical flips, small rotations (±15°), and brightness adjustments were applied during training. This ensures that the model is less sensitive to rotation and illumination variance, addressing the absence of augmentation in earlier experiments.Satellite images taken at two distinct times (T1 and T2) are used in the data preparation step, along with matching labeled masks that describe the changes in land cover. Three directories /T1, /T2, and /label are used to arrange these pictures.All images were resized from their native 512×512 resolution to 256×256 pixels. This reduces computational cost but may also lead to loss of fine boundary details, especially for narrow features such as roads and small buildings. Normalization is applied to the T1 and T2 images.Using a pre-made color map, the RGB images that make up the ground truth masks are transformed into class indices. Based on its RGB value, each pixel in the label image is assigned to a class. Pixels that don't match any predetermined color are remapped to a background class and given an ignore index (255), which removes them from the training loss computation.

B. HRNet Model Architecture

The HRNet framework, on which the model architecture design is based, is especially good at maintaining high-resolution representations across the network. HRNet preserves parallel convolutional branches at various resolutions, in contrast to conventional convolutional networks that downsample features first and then attempt to recover spatial details. After starting with a downsampling stem, the network goes through several stages where features from various resolutions are combined. The network can integrate rich remote sensing information from low-resolution branches with fine spatial details. To create the final segmentation mask with pixel-wise class probabilities, the final feature maps from each branch are upsampled, concatenated, and then run through a convolutional head with a softmax activation

C. Algorithm

The bitemporal satellite images from two different periods (T1 and T2) are inputs for the remote sensing change detection algorithm that uses HRNet. Other inputs include label masks, a designated color map for different classes, target image dimensions, and an ignore index for undefined pixels. The remotely sensed change detection using HRNet model is trained on evaluation metrics such as accuracy, precision, recall, and F1-score which is part of the output for the model.

Each category of land cover is assigned a unique RGB color which makes the visualization clear and consistent in the change detection. Areas with no change between timestamps are colored white (255, 255, 255). Dark green (0, 128, 0) indicates low vegetation, and gray (128, 128, 128) shows surfaces that are barren. Trees are marked with bright green (0, 255, 0) while water bodies are colored blue (0, 0, 255). Buildings and playgrounds are marked with dark red (128, 0, 0) and bright red (255, 0, 0) respectively.

Input:T1_images, T2_images, Label_masks, COLOR_MAP

Output: Trained HRNet model, Evaluation metrics (accuracy,precision, recall, F1-score), Visualized predictions with pixel wise analysis

Steps:

1. Data Preparation (for each image)
   - 1.1. Load image and label mask
   - 1.2. Resize image and mask to (IMG_HEIGHT, IMG_WIDTH)
   - 1.3. Normalize T1 and T2 images to [0, 1]
   - 1.4. Concatenate T1 and T2 images along channel dimension → 6 channels
   - 1.5. Convert label RGB mask to class indices using COLOR_MAP
   - 1.6. Replace undefined pixels with IGNORE_INDEX
   - 1.7. Store processed images in X, labels in Y
2. Dataset Preprocessing
   - 2.1. Split X and Y into training and validation sets (e.g., 80% train, 20% val)
3. Build HRNet Model
   - 3.1. Define input layer with shape (IMG_HEIGHT, IMG_WIDTH, 6)
   - 3.2. Construct stem with convolution, batch norm, ReLU, and downsampling
   - 3.3. Build multiple stages with residual blocks and multi-resolution branches
   - 3.4. Fuse branches at each stage with upsampling/ downsampling and addition
   - 3.5. Build final head to upsample and also output softmax probabilities for the NUM_CLASSES
4. Compile and Train Model
   - 4.1. Compile model with:
     - 4.1.1. optimizer: Adam
     - 4.1.2. loss: categorical cross-entropy
     - 4.1.3. metrics: accuracy
   - 4.2. Train model on training data with validation on validation data for 100 epochs
5. Evaluate Model
   - 5.1. Predict on validation set → Y_pred
   - 5.2. Convert predictions and ground truth to class indices
   - 5.3. Flatten predictions and ground truth for metric calculations
   - 5.4. Compute the metrics like precision, recall, F1-score,accuracy
6. Visualization and Change Summary

   Loop:

   - 6.1. Upload T1, T2, and label images for a test sample
   - 6.2. Load and preprocess uploaded images as in Step 1
   - 6.3. Display T1, T2, and label images side by side
   - 6.4. For the label image, count pixels per class based on COLOR_MAP and display the pixel counts per class

6.5. Use the trained model to predict the change map, visualize the results alongside the original inputs
6.6. Ask user if they want to analyze another image pair; if no, exit loop

7. End Algorithm

D. Training

In the training stage, the model is trained using the categorical cross-entropy loss function, which is suitable for multi-class remote sensing tasks. The Adam optimizer is used for efficient convergence. The dataset is split into 80% training and 20% validation. The model is trained over 100 epochs with a batch size of 4. As the model is evaluated on each class for accuracy, F1-score, recall, and precision, all these metrics are monitored. To prevent overfitting and make the training of the model useful, optimal performance is maintained through early stopping techniques.

TABLE I. Images Details in the Dataset

Label	Value
Image type	RGB
Image actual size	512 x 512 pixels
Classes	6
Image format	.jpg
Training images pairs	2374 (80%)
Testing images pairs	594 (20%)
Total images pairs	2968

As per Table I, The images are of 6 different categories in the dataset, and they are of RGB type having resolution of 512 × 512 pixels. The images are stored in .jpg format. Out of a total of 2,968 image pairs, 2,374 pairs (80%) were used for training, while 594 pairs (20%) for testing.

E. Evaluation

During the evaluation stage, the model is evaluated using performance metrics, such as F1-score, precision, and recall.

TABLE II. Results & Metrics Training

Label	Value
Validation steps	20
Training time	10 to 20 min
Batch size	4
Classification mode	Multiclass Classification
Target size	(256,256,6)
Accuracy	97-98%
Loss	0.15-0.25
Epochs	100

In Table II, important training parameters as well as the results are highlighted.The classification mode was multiclass and a batch size of 4 was used. For the input data, the target size was (256, 256, 6). Over 100 epochs, the model obtained 97-98% accuracy and a loss value of between 0.15 and 0.25.

Accuracy = (True Positives+True Negatives)/Total Samples
Precision = (True Positives+False Positives)/True Positives
Recall = (True Positives+False Negatives)/True Positives
F1-score = 2×(Precision+Recall)/(Precision×Recall)

A classification report that displays the model's performance across all classes is also produced. These metrics offer a thorough understanding of the model's detection and classification capabilities for land changes.

F. Visualization

An interactive user interface is created for the visualization and prediction stage that enables users to upload pairs of bi-temporal remote sensing images (T1 and T2) for the purpose of detecting changes. The original T1, T2, and ground truth images are shown alongside the remote sensing change map produced by the trained HRNet model after processing the inputs. This makes it simpler for users to understand patterns.By facilitating real-time visualization and remote sensing change analysis, the user interface (UI) improves the model's usability and is suitable even for non-technical users.

TABLE III. Performance Comparison to HRNet model

Model	Precision (%)	Recall (%)	F1 Score(%)	Accuracy (%)
U-Net (Baseline)	78.52	76.3	77.39	85.6
DeepLabV3+ (ResNet50)	84.71	81.94	83.3	89.25
HRNet (Our model)	88.86	94.27	91.49	97.08

Table III shows performance on the SECOND dataset. All models were trained under the same conditions (100 epochs, 256×256 input, batch size 4) for fair comparison. Metrics are computed on the test set consisting of 594 image pairs.It shows the strongest performance of HRNet in terms of detecting cover change over time image in comparison to U-Net and DeepLabV3+, SegNet, MobileNetV2+ and DeepLabV3+.The per-class IoU values are Non-change 89.28%, Low vegetation 77.58%, No vegetation surface 85.27%, Tree 86.77%, Building 82.91%, Playground 84.11%, and Water 84.32% with a mean IoU of 84.32%.

HRNet not only achieves higher accuracy but also provides more stable results across all land-cover classes compared to other models. Overall, the combination of high precision, recall, and mean IoU suggests that HRNet is well-suited for real-world remote sensing change detection tasks. Its ability to consistently outperform traditional architectures underlines its potential for scalability and accuracy .

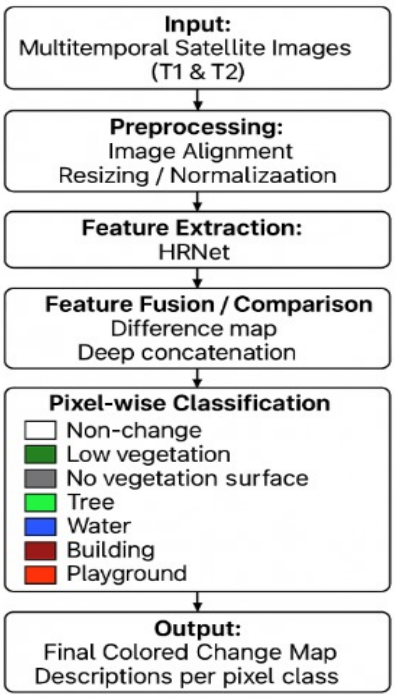

Fig.1 Flowchart

This Fig.1 explains the process of remote sensing change detection on land using HRNet in bitemporal satellite images. It starts with images from two different periods, which undergo a sequence of alignment, resizing, and normalization to achieve pixel-level precision.In the end, each pixel is assigned a class such as vegetation, water bodies, buildings or non-change which results in a change map with land cover pixel wise analysis.

IV. Results and Analysis

This model was evaluated on bi-temporal satellite imagery from the SECOND dataset. It achieved a precision of 88.86%, recall of 94.27%, and F1-score of 91.49%. These results show the capability of the model to detect changes in remote sensing with respect to the various regions so that it shows the effectiveness of the model visualization in presenting the interactive results meaningfully.

To provide an user-friendly experience, a simple web-based Graphical User Interface (GUI) was developed using Streamlit. The interface allows users to upload two satellite images T1 (earlier) and T2 (later) to detect remote sensing land-use changes between them.

Fig.2 Graphical user interface

In Fig.2 GUI allows users to upload T1 and T2 images in JPG, PNG, or JPEG formats,visualize the predicted change map alongside the input images,view pixel-wise remote sensing class statistics for detected land changes. Once the T1 and T2 images are uploaded, the model processes them to generate a remote sensing change map, highlighting land-use transitions such as vegetation loss, building construction, and other land cover changes. The visualization depicts what has changed and how much of each class is present in terms of pixel counts.

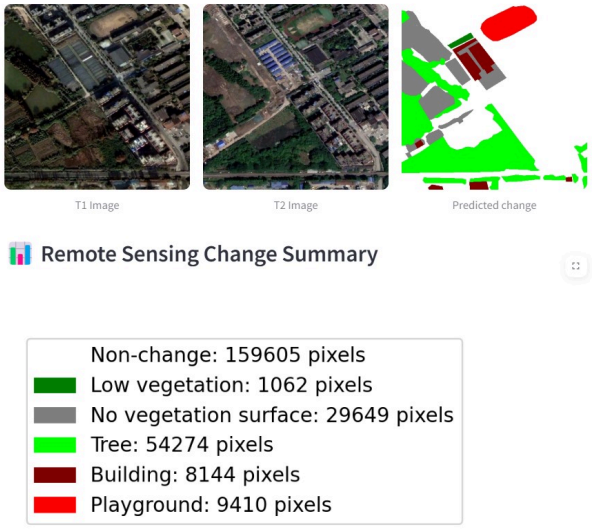

Fig.3 Result 1

As shown in Fig.3, the first example demonstrates significant building activity (8144 pixels) and vegetation changes in low vegetation: 1062 pixels, tree cover: 54274 pixels. The total number of unchanged pixels was 159,605, indicating a substantial portion of the area remained stable across the two time periods.

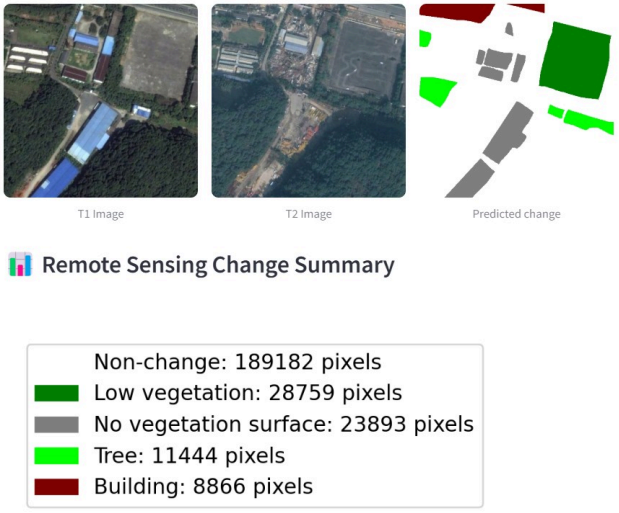

Fig 4 : Result 2

In Fig.4 the second result, a different landscape is observed with major changes in low vegetation (28,759 pixels), bare/no vegetation surfaces (23,893 pixels), and building development (8866 pixels). The model also recorded 189182 unchanged pixels, further emphasizing its capability to accurately distinguish between static and dynamic land regions.

V. Conclusion and Future work

This model implements a change detection method using remote sensing data with a custom deep learning model based on HRNet. The model received pairs of remote sensing images T1 and T2 as training data, alongside pixel-level semantic masks indicating changes in land cover classification: tree, water, building, low vegetation, high vegetation, and playground.Results demonstrated the HRNet model's capability in accurately detecting and segmenting the changes. Evaluated performance metrics like model accuracy, precision, recall, and F1 score on unseen data.The integration of a visualization module and GUI further enhanced interpretability, making the results accessible to both technical and non-technical users. However, the study has several limitations,training was performed on resized 256×256 images, which may reduce boundary precision.No advanced augmentation or post-processing was incorporated, potentially affecting robustness and the computational cost of HRNet remains high, limiting deployment on resource-constrained platforms. Future research will focus on training with full-resolution data, integrating transformer-based temporal attention modules, and developing lightweight HRNet variants. These directions will help strengthen both the accuracy and practicality of HRNet-based change detection for large-scale remote sensing tasks.

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